Chapter 2: C Basics

The C programming language, developed by Dennis Ritchie in the early 1970s, remains one of the most widely used and powerful programming languages. This chapter will introduce you to the fundamental concepts of C, including variables, data types, operators, expressions, and essential input/output functions. Mastering these basics is crucial for understanding more advanced topics in C programming.

2.1 Variables and Data Types

Variables

A **variable** is a named location in memory used to store data. In C, every variable has a specific type that defines the kind of data it can hold. Before using a variable, it must be declared, specifying its type and optionally assigning it an initial value.

Syntax:

data_type variable_name;

Example: